



Modelling Financial Markets

Seminar 2: Market Efficiency — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AMH: Adaptive Markets Hypothesis
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)
- ▣ OLS: Ordinary Least Squares
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SE: Standard Error
- ▣ TOM: Turn Of the Month
- ▣ US: United States
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Key Definitions

Part A: Pen and Paper

Part B: Computation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Compute variance ratios, runs tests and Hurst exponents step by step
- ▣ Test the random walk with inference robust to heteroskedasticity
- ▣ Measure time-varying efficiency with rolling windows and bootstrap bands
- ▣ Test calendar anomalies with HAC errors and multiple-testing corrections

- ▣ Background: Lecture 2; textbooks [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 2 and [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** six pen-and-paper exercises (A1–A6)
- ▣ **Part B:** seven computational exercises with inference (B1–B7)
- ▣ **Part C:** one open question, a seed for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- Report every statistic with its uncertainty (standard error, band or p -value)
- Always compare a classical test with its robust version: when do they disagree, and why?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (variance ratio, step by step), A3 (runs test, step by step), A5 (Hurst exponent from a variance ratio)
- Part B: B1 (homoskedastic vs robust VR), B3 (rolling DFA on the S&P 500), B5 (turn of the month with HAC errors)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (model: A5)
- Part B: B2 (model: B1), B4 (model: B3, B1), B6 (model: B5), B7 (model: B1)
- Part C: C1, open question and project seed; C2, critique an AI answer (model: A1, B1, B7)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter2_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: S&P 500, BET, BET-TR, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai, Euro Stoxx 50, Bitcoin; daily closes up to 18 September 2026
- Equity indices: weekdays only, days with an unchanged close (holidays) removed; crypto keeps all 7 days of the week
- Returns: daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar

Key Definitions (1/2): Random Walk Tests

- **RW3**: uncorrelated increments, volatility may change; the hypothesis tested today
- **Variance ratio** Lo & MacKinlay (1988): $VR(q) = \text{Var}(r_t + \dots + r_{t-q+1}) / (q \text{Var}(r_t)) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$
 - ▶ $z(q)$: assumes constant variance; $z^*(q)$: robust to heteroskedasticity (formulas in Lecture 2)
- **Chow–Denning** Chow & Denning (1993): $\max_i |z^*(q_i)|$ over m horizons; 5% critical value 2.49 for $m = 4$
- **Runs test** Wald & Wolfowitz (1940): number of runs of equal signs R , $\mathbb{E}[R] = 2n_1 n_2 / n + 1$, $\text{Var}[R] = 2n_1 n_2 (2n_1 n_2 - n) / (n^2 (n - 1))$
- **Hurst exponent** H : scaling of the range, $0.5 = \text{no memory}$; DFA exponent Peng et al. (1994); self-similar case $VR(q) = q^{2H-1}$

Key Definitions (2/2): Inference Tools

- ▣ **Robust SE of $\hat{\rho}_k$:** $\sqrt{\hat{\tau}_k}$, $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$, with $e_t = r_t - \bar{r}$
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) errors [Newey & West \(1987\)](#): valid when residuals are heteroskedastic and autocorrelated
- ▣ **Wild bootstrap** [Kim \(2006\)](#): multiply centred returns by random signs ± 1 ; keeps the volatility path, destroys autocorrelation
- ▣ **Block bootstrap** [Künsch \(1989\)](#): resample blocks of consecutive days; keeps short-run dependence
- ▣ **FWER**: probability of at least one false rejection; **Bonferroni**: $p_i < \alpha/M$; **Holm**: sorted $p_{(j)} < \alpha/(M - j + 1)$, stop at the first failure

A1: Variance Ratio, Step by Step [Solved]

Task

- Eight daily returns (%):
1.0, -0.5, 0.8, 0.3, -1.2, 0.6, 0.9, -0.4
- (a) Mean $\hat{\mu}$ and variance $\hat{\sigma}_1^2 = \frac{1}{T} \sum (r_t - \hat{\mu})^2$
- (b) Overlapping 2-day sums and $\hat{\sigma}_2^2 = \text{mean of } (s_t - 2\hat{\mu})^2$
- (c) $\widehat{VR}(2) = \hat{\sigma}_2^2 / (2\hat{\sigma}_1^2)$; compare with $1 + \hat{\rho}_1$

Solution

- $\hat{\mu} = 0.1875\%$, $\hat{\sigma}_1^2 = 0.5586$
- Sums: 0.5, 0.3, 1.1, -0.9, -0.6, 1.5, 0.5; $\hat{\sigma}_2^2 = 0.6292$
- $\widehat{VR}(2) = 0.6292 / (2 \cdot 0.5586) = 0.563$
- $\hat{\rho}_1 = -0.395$, so $1 + \hat{\rho}_1 = 0.605$: close; the gap comes from the end effects of a tiny sample
- $VR < 1$: strong mean reversion (signs alternate)

A2: Variance Ratio at Three Days [Proposed]

Solution discussed in class

Task

- Same eight returns as A1; now $q = 3$
- (a) Overlapping 3-day sums and $\hat{\sigma}_3^2$; (b) $\widehat{VR}(3)$
- (c) Compute $\hat{\rho}_2$ and the approximation $1 + 2(\frac{2}{3}\hat{\rho}_1 + \frac{1}{3}\hat{\rho}_2)$
- Why can a real test never be based on 8 observations?
- Model: A1



A3: Runs Test, Step by Step [Solved]

Task

- ▣ Signs of 20 daily returns: ++-+++--+-+---+----
- ▣ (a) Count n_1 , n_2 and the runs R
- ▣ (b) $\mathbb{E}[R]$, $\text{Var}[R]$, z and the two-sided p -value
- ▣ (c) Decide at 5% Wald & Wolfowitz (1940)

Solution

- ▣ $n_1 = 12$, $n_2 = 8$, $R = 11$
- ▣ $\mathbb{E}[R] = 2 \cdot 12 \cdot 8/20 + 1 = 10.60$; $\text{Var}[R] = 4.345$
- ▣ $z = (11 - 10.60)/\sqrt{4.345} = 0.19$, $p = 0.85$
- ▣ Do not reject independence: the number of runs is what chance produces

A4: A Sequence with Momentum [Proposed]

Solution discussed in class

Task

- ▣ Signs: +++++---+++++---
- ▣ (a) n_1 , n_2 , R ; (b) z and p ; (c) which autocorrelation sign does this suggest?
- ▣ (d) Would the same signs with very different magnitudes change the test? Why is that a strength and a weakness?
- ▣ Model: A3



A5: Hurst Exponent from a Variance Ratio [Solved]

Task

- ▣ For self-similar increments $VR(q) = q^{2H-1}$
- ▣ (a) Derive H as a function of $VR(q)$ and q
- ▣ (b) BET, 2000–2026: $\widehat{VR}(20) = 1.377$; compute $\hat{H}$
- ▣ (c) Compare with the corrected R/S estimate 0.533 of Lecture 2

Solution

- ▣ $\ln VR(q) = (2H - 1) \ln q \Rightarrow H = 0.5 + \ln VR(q) / (2 \ln q)$
- ▣ $\hat{H} = 0.5 + 0.3199 / 5.991 = 0.553$
- ▣ Close to the R/S value: both say “mild persistence”
- ▣ Caveat: short memory (an AR(1) with $\rho_1 = 0.109$) gives the same VR; H from VR does not prove long memory

A6: Hurst Exponent of the S&P 500 [Proposed]

Solution discussed in class

Task

- ▣ S&P 500, 2000–2026: $\widehat{VR}(20) = 0.753$
- ▣ (a) Compute $\hat{H}$; (b) persistent or anti-persistent?
- ▣ (c) For an AR(1) with $\rho_1 = -0.1$, compute $VR(20)$ from $1 + 2 \sum_k (1 - k/q) \rho_1^k$ and the implied H : what do you conclude?
- ▣ Model: A5

B1: Homoskedastic vs Robust Variance Ratio [Solved]

Task

- ▣ S&P 500 and BET, daily log returns 2000–2026
- ▣ (a) $\widehat{VR}(q)$, $z(q)$ and $z^*(q)$ for $q = 2, 5, 20$ Lo & MacKinlay (1988)
- ▣ (b) Chow–Denning over $q = 2, 5, 10, 20$ Chow & Denning (1993)
- ▣ Code: `variance_ratio`, `chow_denning` in the notebook

Interpretation

- ▣ Why is $|z| > |z^*|$ in both markets?
- ▣ Does the sign of $VR - 1$ agree with $\hat{\rho}_1$ from the lecture?
- ▣ Would you trade on the S&P 500 reversal? What is missing from the test?

B1: Results [Solved]

	$\widehat{VR}(2)$	$z(2)$	$z^*(2)$	$\widehat{VR}(5)$	$z(5)$	$z^*(5)$	$\widehat{VR}(20)$	$z^*(20)$
S&P 500	0.901	-8.12	-3.64	0.830	-6.37	-2.77	0.753	-1.86
BET	1.108	8.83	3.84	1.173	6.44	2.98	1.377	3.31

- Chow–Denning: S&P 500 3.64 ($p = 0.0011$), BET 3.84 ($p = 0.0005$): both reject RW3
- z is two to three times z^* : volatility clustering makes $\text{Var}(\hat{\rho}_k) > 1/T$
- Signs agree with $\hat{\rho}_1$: reversal for the S&P 500, persistence for BET
- Missing: trading costs, the joint-hypothesis problem and stability over time (B3, B7)

B2: BET-TR and Emerging Markets [Proposed]

Solution discussed in class

Task

- Repeat B1 for BET-TR (2014–2026) and WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai
- Report $\widehat{VR}(2)$, $z(2)$, $z^*(2)$, $z^*(5)$ and the Chow–Denning p -value
- Model: B1

Interpretation

- How many markets reject with $z(2)$, and how many with Chow–Denning?
- BET-TR includes dividends: why can it differ from BET? Is its sample comparable?
- Are emerging markets less efficient as a group? What else differs (liquidity, size, trading hours)?

B3: Rolling DFA on the S&P 500 [Solved]

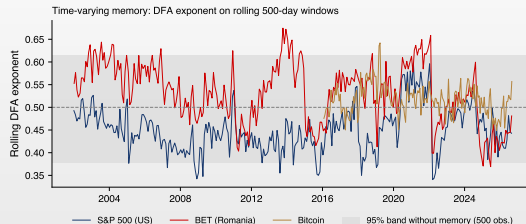
Task

- (a) DFA exponent on rolling 500-day windows, step 21 days [Peng et al. \(1994\)](#)
- (b) Null band: DFA of 300 simulated i.i.d. Student- t_4 series of length 500; keep the 2.5% and 97.5% quantiles
- (c) Share of windows outside the band; dates of the extremes

Interpretation

- Is the share of windows outside the band larger than 5%?
- Why Student- t and not the Normal distribution for the null?
- Why is the band so much wider than the full-sample band of Lecture 2?

B3: Results [Solved]



- Null band $[0.378, 0.614]$; windows outside: 6.4%, close to the 5% expected by chance
- Mean exponent 0.450; minimum 0.341 (window ending 2022-03-15), maximum 0.598 (2021-02-12)
- The mean below 0.5 matches the negative $\hat{\rho}_1$: mild anti-persistence, not long memory
- Student- t_4 : heavy tails widen the sampling distribution of the DFA exponent; the Normal distribution would give a too narrow band
- 500 observations instead of 6 714: the standard error grows roughly with $1/\sqrt{T}$

B4: Adaptive Efficiency of Bitcoin [Proposed]

Solution discussed in class

Task

- Bitcoin from 2011; rolling 365-day windows, step 30 days [Lo \(2004\)](#)
- (a) $z^*(5)$ in each window
- (b) Wild-bootstrap band per window [Kim \(2006\)](#): 99 replications of $e_t \cdot \eta_t$, $\eta_t = \pm 1$ with probability $1/2$
- (c) Windows outside the band; compare with [Urquhart \(2016\)](#)
- Model: B3 (rolling windows), B1 (z^*)

Interpretation

- Is Bitcoin becoming more efficient over time, as the AMH suggests?
- Why is the bootstrap band not symmetric and not equal to ± 1.96 ?
- With about 180 windows, how many rejections would pure chance produce?

B5: The Turn-of-the-Month Effect with HAC Errors [Solved]

Task

- S&P 500 and BET, 2000–2026, daily returns in bp
- $TOM_t = 1$ on the last trading day of a month and the first three of the next [Ariel \(1987\)](#)
- Regress $r_t = a + b TOM_t + u_t$; report $\hat{b}$ with OLS and with HAC (5 lags) standard errors [Newey & West \(1987\)](#)

Interpretation

- Why do the OLS and HAC standard errors differ?
- Is $+\hat{b}$ bp per day large? Compute it per month (4 TOM days)
- Which explanations would you test next (flows, pension funds, index rebalancing)?

B5: Results [Solved]

	TOM share	$\hat{a}$ (bp)	$\hat{b}$ (bp)	SE_{OLS}	t_{OLS}	SE_{HAC}	t_{HAC} (p)
S&P 500	19.1%	1.8	3.3	3.8	0.88	3.5	0.95 (0.34)
BET	19.3%	3.0	17.8	4.3	4.11	4.8	3.74 (0.0002)

- S&P 500: TOM days earn 5.2 bp vs 1.8 bp on other days: not significant
- BET: 20.8 bp vs 3.0 bp; about $4 \times 17.8 \approx 71$ bp extra per month
- BET: the HAC error is larger than the OLS error (volatility clustering, $\rho_1 > 0$), so t falls from 4.11 to 3.74; still significant
- One market, one effect: before believing it, correct for the number of effects tried (B6)

 MFM_ch2_calendar_anomalies

B6: Fifteen Calendar Tests and Multiple Testing [Proposed]

Solution discussed in class

Task

- ▣ Markets: S&P 500, BET, Euro Stoxx 50, WIG20, BUX; effects: day of the week (Wald test of equal means), January (monthly returns), turn of the month; all with HAC errors
- ▣ (a) The 15 p -values; (b) raw decisions at 5%; (c) Bonferroni and Holm decisions
- ▣ Model: B5

Interpretation

- ▣ What is the FWER of 15 independent tests at 5%?
- ▣ Which effect is “significant” only before the correction? Would you publish it?
- ▣ [Harvey, Liu & Zhu \(2016\)](#) ask for $|t| > 3$: which effects pass this bar?

B7: Is the S&P 500 Reversal a Crisis Effect? [Proposed]

Solution discussed in class

Task

- S&P 500, 2000–2026; remove Sep 2008 – Jun 2009 and 15 Feb – 30 Jun 2020
- (a) $\hat{\rho}_1$ with the i.i.d. $t = \hat{\rho}_1\sqrt{T}$ and the robust $t = \hat{\rho}_1/\sqrt{\hat{\tau}_1}$, with and without the crisis days
- (b) $z^*(5)$ in both samples
- Model: B1

Interpretation

- How much of the full-sample autocorrelation comes from 14 crisis months?
- Why does the gap between i.i.d. and robust t shrink without crises?
- Is removing crisis days legitimate? When is it data snooping?

C1: Did the BVB Become More Efficient after the 2020 FTSE Upgrade? [Proposed]

Solution discussed in class

The question

- **FTSE Russell (2019)**: Romania moved from Frontier to Secondary Emerging on 21 September 2020
- More foreign investors and analysts should speed up price discovery: did the BET become closer to a random walk?
- Background on the BVB: **Pele & Mazurencu-Marinescu (2012)**; this question extends Seminar 1, C1

Deliverable

- A Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should contain

- $\hat{\rho}_1$ and $z^*(5)$ for five years before and after the event
- A block-bootstrap confidence interval for the change in $\hat{\rho}_1$ **Künsch (1989)**
- A rolling view (B3) and a comparison group (WIG20, BUX) to separate the event from common shocks
- Confounders: COVID-19, index composition (Hidroelectrica listed in 2023), liquidity
- The link with the Adaptive Markets Hypothesis **Lo (2004)**

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “Test whether the S&P 500 (daily closes, 2000–2026) follows a random walk with the variance ratio. Give Python code and interpret the result”
- **The answer:** the code below, followed by the interpretation

```
r = np.log(sp).diff().dropna()           # daily log returns
T = len(r)
def vr(q):
    rq = r.rolling(q).sum().dropna()      # overlapping q-day returns
    return rq.var() / (q * r.var())
q = 5
z = (vr(q) - 1) / np.sqrt(2 * (2*q - 1) * (q - 1) / (3 * q * T))
p = 2 * stats.norm.sf(abs(z))
```

- **Interpretation of the AI**
 - ▶ “VR(2) = 0.901, VR(5) = 0.830, $z(5) = -6.37$, $p < 0.001$: the random walk is rejected”
 - ▶ “Since VR(2) = $1 + 2\rho_1$, the first-order autocorrelation is $\rho_1 \approx -0.050$ ”
 - ▶ “VR < 1 means positive autocorrelation, i.e. momentum: buy the S&P 500 after up days”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- (a) Find the errors: there are two or three; the rest of the answer is correct
- (b) For each error, say how you would detect it: a test, a published number, the lecture or the library documentation
- (c) Give the corrected code and the corrected interpretation
- (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- Model: A1, B1 and B7

Why this exercise

- AI code usually runs without an error message: the mistakes are in the statistics, not in the syntax
- Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

Common Pitfalls

- ▣ Using $\pm 1.96/\sqrt{T}$ bands or homoskedastic z with volatility clustering: too many rejections
- ▣ Reading the best of many horizons, markets or calendar effects without a multiple-testing correction
- ▣ Taking an uncorrected R/S slope above 0.5 as proof of long memory
- ▣ Counting overlapping rolling windows as independent evidence
- ▣ Equating statistical rejection with a profitable strategy: costs and risk are missing
- ▣ Filling holidays with the previous close: creates zero returns and fake runs

Take-Home Assignment (Team Project)

- ▣ Choose one market not covered in class (another index, an exchange rate or a crypto-asset)
- ▣ Run the full battery: robust $\hat{\rho}_k$, $\tilde{Q}(10)$, runs, $z^*(q)$, Chow–Denning, Lo's V , DFA with shuffled band
- ▣ Add a rolling analysis with a proper null band and one calendar effect with HAC errors and Holm correction
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_02

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